	maximum height.
3(a)(i)	For Geostationary Satellite, period of orbit = 24 hrs = 86400 s $F_c = mr\omega^2$ $F_c = mr\left(\frac{2\pi}{T}\right)^2$ $F_c = (6800)\left[(6400+36000)\times 10^3\right]\left(\frac{2\pi}{86400}\right)^2$ $F_c = 1524 \approx 1500 \text{ N}$
3(a)(ii)	For a satellite in circular orbit, the centripetal force is supplied by the gravitational force acting on the satellite, $F_c = F_G$ $mr\left(\frac{2\pi}{T}\right)^2 = \frac{GMm}{r^2}$ $T = \sqrt{\frac{4\pi^2 r^3}{GM}}$ From the above equation, for a geostationary orbit (period = 24 Hr), the mass of the satellite is not a fixed value, and hence, there is no requirement for it to be 6800 kg.
3(b)(i)	Since motion of ion is perpendicular to the magnetic field, $F_B = F_c$ $Bqv = \frac{mv^2}{r}$ $Bq = \frac{mv}{r}$ $r = \frac{mv}{Bq}$ $r = \frac{3.4\times 10^{-23}}{(1.8\times 10^{-3})(1.6\times 10^{-19})}$ $r = 0.118 \approx 0.12 \text{ m}$
3(b)(ii)	The radius of the circular path in this magnetic field depends on the charged particle's momentum and charge. For ions with the same charge but different momentum, its radius will be different.
4(a)(i)	Incident sound wave from the signator is reflected off the metal plate. There are